

CHAPTER 6

IMPLEMENTATION AND OUTPUT

IMPLEMENTATION:

1. **Encrypt Text:** Generate a key and IV, encrypt plaintext, and display results in Hex/Base64.
2. **Decrypt Text:** Provide ciphertext, key, and IV to retrieve the original plaintext.
3. **Encrypt Files:** Compress and encrypt files, optionally save key/IV.
4. **Decrypt Files:** Decrypt and decompress files to retrieve original content.
5. **Encrypted Chat:** Secure real-time communication with automatic encryption for messages.
6. **Help & Exit:** View usage instructions or exit the tool.

OUTPUT:

```
=====
      AES Encryption/Decryption Tool
=====
1. Encrypt Text
2. Decrypt Text
3. Encrypt/Decrypt File
4. Chat Application
5. Help
6. Exit
Enter choice (1/2/3/4/5/6): █
```

FIG 6.1 MAIN INTERFACE

18

```
=====
      AES Encryption/Decryption Tool
=====
1. Encrypt Text
2. Decrypt Text
3. Encrypt/Decrypt File
4. Chat Application
5. Help
6. Exit
Enter choice (1/2/3/4/5/6): 1

=====
      Encryption
=====
Enter plaintext to encrypt: HELLO
Encrypting...

Encryption Details:
Encryption Key (Hex): 10ac74b87c#1db7e349aa6eda92ff64f
Encryption IV (Hex): a52c3517c5f52f93faab8b942bbeceb4
Ciphertext (Hex): a52c3517c5f52f93faab8b942bbeceb48ccb2345fad7ea6fedbd4d1774acc177
Ciphertext (Base64): p5e1F8X1L5P664uUK770tIvI.DX61+pv7b1NF3RPaXc=
```

FIG 6.2 TEXT ENCRYPTION

```

=====
AES Encryption/Decryption Tool
=====
1. Encrypt Text
2. Decrypt Text
3. Encrypt/Decrypt File
4. Chat Application
5. Help
6. Exit
Enter choice (1/2/3/4/5/6): 2

=====
Decryption
=====
Enter ciphertext (Base64) to decrypt: p5w1F8X1L5P664uHk770tIvLI0X61-pv7b1NF3RwXc=
Enter encryption key (Hex): 10ec74b87cf1db7e349aa4eda92ffe4f
Enter encryption IV (Hex): a52c3517c5f52f93faeb8b942bbeceb4
Decrypting...

Decryption Result:
Decrypted Plaintext: HELLO

```

FIG 6.3 TEXT DECRYPTION

19

```

=====
AES Encryption/Decryption Tool
=====
1. Encrypt Text
2. Decrypt Text
3. Encrypt/Decrypt File
4. Chat Application
5. Help
6. Exit
Enter choice (1/2/3/4/5/6): 3

=====
File Encryption/Decryption
=====
Encrypt or decrypt file? (e/d): e
Enter file path: C:\Downloads\Programming\Python\AES Tool\Sample_Text.txt
Enter path to save key and IV (or press Enter to skip): C:\Downloads\Programming\Python\AES Tool\Save_iv_key.txt
Encrypting file...
File encrypted and saved as C:\Downloads\Programming\Python\AES Tool\Sample_Text.txt.enc

```

FIG 6.4 FILE ENCRYPTION

```

=====
AES Encryption/Decryption Tool
=====
1. Encrypt Text
2. Decrypt Text
3. Encrypt/Decrypt File
4. Chat Application
5. Help
6. Exit
Enter choice (1/2/3/4/5/6): 3

=====
File Encryption/Decryption
=====
Encrypt or decrypt file? (e/d): d
Enter file path: C:\Downloads\Programming\Python\AES Tool\Sample_Text.txt.enc
Enter encryption key (Hex): ede0633a204663db44319a1dc0d1139c
Enter encryption IV (Hex): 682c4b724488a5c5bc9b2ada37d6d593
Decrypting file...
File decrypted and saved as C:\Downloads\Programming\Python\AES Tool\Sample_Text.txt_decrypted.enc

```

FIG 6.5 FILE DECRYPTION

```

=====
Chat Application
=====
Enter your encryption key (Hex): ede0633a204663db44319a1dc0d1139c
Start server or client? (s/c): s
Server started! Waiting for connections...
Accepted connection from ('127.0.0.1', 53155)
Type 'exit' to shut down the server: Received: Hello

```

FIG 6.6 CHAT APPLICATION – SERVER SIDE

20



```

=====
Chat Application
=====
Enter your encryption key (Hex): ede0633a204663db44319a1dc0d1139c
Start server or client? (s/c): s
Server started! Waiting for connections...
Accepted connection from ('127.0.0.1', 53155)
Type 'exit' to shut down the server: Received: Hello

```

FIG 6.6 CHAT APPLICATION – SERVER SIDE

20

```

=====
Chat Application
=====
Enter your encryption key (Hex): ede0633a204663db44319a1dc0d1139c
Start server or client? (s/c): c
Enter message to send (or 'exit' to quit): Hello
Encrypted Message (Hex): f5d4a3bd990cdd0aab92f4a48ccbd6764e74de3fe2f345982744
9e83fab4cce9
Encrypted Message (Base64): 9dSjvZkM3QgrkvSkjMvWdkS03j/i8OWQJ8Seg/q8z0k=
Enter message to send (or 'exit' to quit):

```

FIG 6.7 CHAT APPLICATION – CLIENT SIDE

```

=====
AES Encryption/Decryption Tool
=====
1. Encrypt Text
2. Decrypt Text
3. Encrypt/Decrypt File
4. Chat Application
5. Help
6. Exit
Enter choice (1/2/3/4/5/6): 6

=====
Exiting...
=====

```

FIG 6.8 EXITING/TERMINATING

21